

Assignment no 3

Code explanation-

- **Import libraries** → Load required Python packages.
- **Load dataset** → Load Diabetes dataset into pandas DataFrame.
- **Quick look** → Check first rows, stats, and missing data.
- **EDA** → Plot histograms and correlation heatmap.
- **Stratified Sampling** → Split dataset into training/testing sets while maintaining target distribution.
- **Separate features/target** → Prepare X and y for modeling.
- **Feature Scaling** → Standardize features for models sensitive to scale.
- **Linear Regression** → Train/test linear regression.
- **Ridge Regression** → Train/test linear regression with L2 regularization.
- **Polynomial Regression** → Train/test degree 3 & 5 polynomial regression.
- **Nonlinear Regression (SVR)** → Train/test Support Vector Regression.
- **Ensemble Method** → Train/test Random Forest Regressor.

Results and discussion-

9) RMSE Measurement

- RMSE computed using mean_squared_error:
 - Linear: 54.54
 - Ridge: 54.27
 - Polynomial (deg 3): 210.61
 - Polynomial (deg 5): 295.84
 - SVR: 51.89
 - Random Forest: 53.23

10) Performance of Models

Model	Performance
Linear Regression	Moderate performance, slight underfitting
Ridge Regression	Moderate performance, slight underfitting
Polynomial Regression	Extreme overfitting at high degrees

Model	Performance
SVR (Support Vector Regressor)	Best nonlinear performance, generalizes well
Random Forest Regressor	Good performance, balances bias-variance, handles nonlinearity

11) Linear vs Nonlinear Models

- Linear models are straightforward and easy to interpret, but they struggle to capture more complex, nonlinear relationships in the data.
- Nonlinear models like SVR and Random Forest can handle these complex patterns much better and provide more accurate predictions.
- Polynomial regression tends to overfit when the degree is too high, whereas SVR and Random Forest manage to generalize well to unseen data.

12. Approaches to Improve Performance

- Adjust the model settings carefully (like using GridSearchCV) to make the models work better.
- Focus on the important features and remove any unnecessary or noisy ones.
- Check the model's performance on different sets of data using cross-validation to be sure it will work well in real situations.
- For polynomial models, keep the degree lower or use regularization to avoid overfitting.
- Combine multiple models together (ensembles or stacking) to get more accurate and reliable predictions.